CSC311 Introduction to Machine Learning   Week 1 Exercises

These homework exercise questions are intended to help you learn the CSC311 course materials, while also preparing you for the term tests. There are three types of questions: the "Training" questions are intended to be straightforward applications of the course materials; start here so that you develop a solid understanding of the fundamental ideas. The "Validation" questions are intended to check whether you can apply the ideas and concepts to new settings. Finally, the "Generalization" questions stretch a little beyond what we cover directly, but are intended to challenge you and expand your understanding of the materials. You are welcome to attempt these questions if you are confident in your understanding of the fundamental ideas.

Every week, you may submit up to 3 pages[1] of solutions to receive TA feedback. You should choose questions that are just challenging enough to you that you feel like feedback would be helpful. Even if you believe that you are able to solve a problem, you might find it helpful to write out the solution. It can help to receive feedback on your ability to communicate your solutions.

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

We will not be posting solutions to these problems. However, you are welcome to discuss solutions with other students and with the TAs/instructors during office hours. You will also receive feedback from TAs on the solutions that you submit.

Collaboration is permitted. We encourage you to find a study strategy that work for you, keeping in mind that tests are done individually.

Training

W1-T1. What is the difference between supervised learning, unsupervised learning, and reinforcement learning? Why is a reinforcement learning problem "harder" than a supervised learning problem?

W1-T2. Draw the decision boundary of a 1NN model trained on the below data set. Label the predictions in each region.

$$\mathbf{X} = \begin{bmatrix} 2 & 1 \\ 2 & 2 \\ 1 & 5 \\ 2 & 4 \\ 4 & 3 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

W1-T3. What is the purpose of the training, validation, and test sets? What can happen to the training, validation, and test error if k (In k nearest neighbours) is too small? What about too large?

W1-T4. What do the terms "overfitting" and "underfitting" mean? How does the choice of k impact a kNN model's chances of overfitting or underfitting? How does the size of the training data set affect a trained model's propencity to overfit or underfit?

W1-T5. Explain why normalizing the features in your data set prior to training a kNN model can improve the model's performance.

W1-T6. Which of these statements about the curse of dimensionality are true?

(a) The curse of dimensionality states that if we would like the nearest neighbour of any query $\mathbf{x} \in \mathbb{R}^d$ to be at least ϵ distance away from a training data point, then we will need $O(d^2)$ data points in our model.

(b) If the training data is uniformly distributed in high dimensions, we would expect a query (input data point) $\mathbf{x} \in \mathbb{R}^d$ to be similarly close to every training data point.

(c) Normalization alleviates (reduce the impact of) the curse of dimensionality.

[1] may change depending on uptake and TA grading time.

W1-T7. Chansey would like to build a machine learning agent that suggests which folders a CSC311 Piazza question should be placed. When a new question is posted, the model would suggest whether the question should be placed in the folder 'lectures', 'tutorials', 'homework', 'tests', or 'other'. This model will be deployed during the next year's CSC311 offering.

Chansey has, as training data, the past 2 years of CSC311 Piazza questions, including the question text, the question asker id, the answer text, the question answerer id, and the folder where the question appeared. Although instructors used different folder names over the years, the folders map to the 5 categories that were mentioned.

(a) Is this a supervised, unsupervised, or reinforcement learning problem? Is this a regression, classification, clustering, or other type of problem?

(b) Suppose Chansey has 700 Piazza questions across the past 2 years of CSC311 Piazza questions. Chansey decides to split the Piazza questions randomly: 420 data points will be used for training (60%), 140 for validation (20%) and 140 for testing (20%). The decision about which set a data point belongs to is made independently of any other data point. Explain why this split may cause Chansey to **over-estimate** the test accuracy. Provide an alternative approach to create the test set that would avoid this issue.

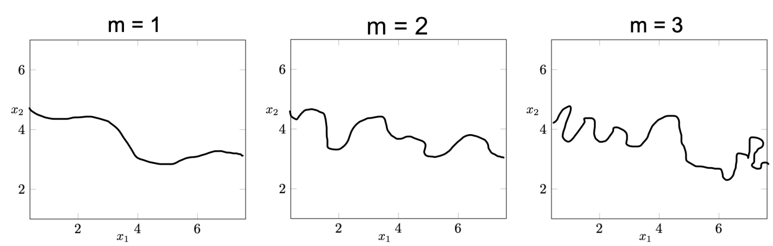
(c) Chansey considers several models when building a solution, including a nearest-neighbour model and a linear model. Chansey knows that we will need the validation set to determine the optimal k in the nearest-neighbour model, but which data set should Chansey use to decide betwen the kNN model and the linear model? Explain your choice.

Validation

W1-V1. We saw, during lecture, that when Euclidean distances are used, kNN is sensitive to data normalization. That is, normalizing the features changes the prediction made by the kNN model. Is kNN still sensitive to data normalization if *cosine similarity* is used as the distance metric? If your answer is "yes", show that this is the case using a simple example (i.e., create a data set and demonstrate the difference). If your answer is "no", prove it.

W1-V2. Explain why the 1NN model will generally have a training accuracy of 100%. In what situation would the training accuarcy be exactly 100%?

W1-V3. Suppose that you are given a mystery model with a hyperparameter m, and this mystery model can be trained to perform binary classification. You train this model on a data set with features x_1 and x_2 using three different hyperparameters settings, and observe the following decision boundaries:



(a) Is it true that increasing m likely increases the model's training accuracy? Justify your reasoning in 1-2 sentences.

(b) Lisa believes that if we had a larger training set, the decision boundary of our model when $m = 3$ will look smoother. What assumption is she making?

Generalization

W1-G1. We saw that the decision boundary of a 1-nearest neighbour consists of several line segments. What about a 3-nearest neighbour model? Argue that the decision boundary of a 3NN model also consists of several line segments. Will these line segments tend to be longer or shorter than those produced by 1-nearest neighbour?

W1-G2. Suppose $\mathbf{x} = (x_1, x_2, \ldots, x_d)$ is a random vector where each x_j is an independent uniform random variable on $[0, 1]$.

(a) Derive the probability that all coordinates of $\mathbf{x}$ lie within $[0, 0.1]^d$. That is, the derive the probability that $||\mathbf{x}||_\infty \leq 0.1$, where $||\cdot||_\infty$ is the L_∞ norm and is defined as follows: $||\mathbf{x}||_\infty = \max_j x_j$.

(b) Evaluate this probability for $d = 2, 5, 10, 50$. What happens as d increases? How does this result demonstrate the curse of dimensionality?

W1-G3. Suppose that $\mathbf{x}$ and $\mathbf{y}$ are two independent random vectors, where each coordinate x_j and y_j is drawn independently and uniformly from $[0, 1]$. Define the squared Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$ as:

$$\|\mathbf{x} - \mathbf{y}\|_2^2 = \sum_{j=1}^{d} (x_j - y_j)^2.$$

(a) Show that the expected squared distance is $\mathbb{E}[\|\mathbf{x} - \mathbf{y}\|_2^2] = \frac{d}{6}$. Hint: You will need to integrate over x_j and y_j using the probability density.

(b) Show that the variance of $\|\mathbf{x} - \mathbf{y}\|_2^2$ is $d\frac{7}{180}$.

(c) The coefficient of variation (CV) is defined as the ratio of the standard deviation to the mean. Show that the CV of $[\|\mathbf{x} - \mathbf{y}\|_2^2]$ decreases with d. What does this tell you about the relative differences between distances in high-dimensional spaces?

W1-G4. Suppose we have high-dimensional data that is not evenly distributed. Can you give an example of circumstances under which the curse of dimensionality would not apply?

W1-G5. For some models, training may be non-deterministic or sensitive to small changes in hyperparameters. In such cases, we may get different models with different accuracies even if we change very little about their training. Suppose we have a model with hyperparameter k. k does nothing of significance, but if we tune k with a validation set, we will get a slightly, randomly different model for every value of k.

Sometimes, researchers do not use a test set and only report their validation accuracy. Based on the above paragraph, why might this be problematic?